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Sequencing hour-level temporal patterns of polysubstance use among persons who use cocaine, alcohol, and cannabis: A back-translational approach

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ABSTRACT

Background: Polysubstance use is highly prevalent among persons who use cocaine; however, little is known about how alcohol and cannabis are used with cocaine. We identified temporal patterns of cocaine+alcohol and cocaine+cannabis polysubstance use to inform more translationally relevant preclinical models.

Methods: Participants who used cocaine plus alcohol and/or cannabis at least once in the past 30 days (n=148) were interviewed using the computerized Substance Abuse Module and the newer Polysubstance Use–Temporal Patterns Section. For each day in the past 30 days, participants reported whether they had used cocaine, alcohol, and cannabis; if any combinations of use were endorsed, participants described detailed hourly use of each substance on the most "typical day" for the combination. Sequence analysis and hierarchical clustering were applied to identify patterns of timing of drug intake on typical days of cocaine polysubstance use.

Results: We identified five temporal patterns among the 180 sequences of reported cocaine polysubstance use: 1) limited cocaine/cocaine+alcohol use (53%); 2) extensive cannabis then cocaine+alcohol+cannabis use (22%); 3) limited alcohol/cannabis then cocaine+alcohol use (13%); 4) extensive cocaine+cannabis then cocaine+alcohol+cannabis use (4%); and 5) extensive cocaine then cocaine+alcohol use (8%). While drug intake patterns differed, prevalence of use disorders did not.

Conclusions: Patterns were characterized by cocaine, alcohol, and cannabis polysubstance use and by the timing, order, duration, and quantity of episode-level substance use. The identification of real-world patterns of cocaine polysubstance use represents an important step toward developing laboratory models that accurately reflect human behavior.

1. Introduction

Cocaine use disorder remains a substantial public health concern. After years of declining rates, cocaine use has been increasing (Jalal et al., 2018; Mustaquim et al., 2021); in 2020, among people aged 12 and older in the US, approximately 5.2 million reported past-year cocaine use, of which 1.3 million met criteria for cocaine use disorder (SAMHSA, 2021). Despite decades of research and development, however, no pharmacotherapies have been approved by the US Food and Drug Administration to treat cocaine use disorder (Bentzley et al., 2021). One explanation for this failure is that most existing animal models that test the efficacy of pharmacotherapies for cocaine use disorder do not reflect real-world patterns of cocaine use.

Most people who use cocaine also use other substances. A meta-analysis estimated that, among those who use cocaine, 74% used alcohol and 38% used cannabis simultaneously with cocaine (Liu et al., 2018). There has been increased recognition of the importance of studying polysubstance use, including the distinction between concurrent use (two or more substances used on separate occasions), sequential use (two or more substances used in a specific temporal order on the

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same occasion), and simultaneous use (two or more substances used at the same time or in close temporal proximity) (Liu et al., 2018). The few studies that have examined the temporal ordering of cocaine use with alcohol and/or cannabis have done so using broad timescales that ask participants to order cocaine use with alcohol and cannabis use for each day of co-use (Gossop et al., 2006), the most recent day of co-use (Barrett et al., 2006), or the most typical day of use (Staines et al., 2001). Such ordering of substance use does not provide sufficient information concerning timing, duration, or temporal proximity of polysubstance use to enable determination of the pharmacological and neurobiological consequences of co-use.

More detailed data regarding the temporal patterns of alcohol or cannabis with cocaine is critical because these patterns can affect cocaine's metabolic profile (McCance-Katz et al., 1998; Pan and Hedaya, 2000; Reid and Bornheim, 2001), potentially enhance its reinforcing effects (Hart et al., 2000; Jatlow et al., 1991; Raven et al., 2000), and reduce the efficacy of anti-relapse therapeutics (Green et al., 2012; Hersh et al., 1998; Schmitz et al., 2009, 2004; Stennett et al., 2020). The presence of alcohol and cocaine in the liver permits production of the psychoactive metabolite cocaethylene, which has a longer half-life than cocaine (Pan and Hedaya, 2000) and equipotency at the dopamine transporter (Jatlow et al., 1991). In the presence of alcohol, cocaine metabolism is also altered to increase plasma concentrations of cocaine (Farré et al., 1997, 1993). Thus, simultaneous use of cocaine and alcohol results in higher brain and plasma concentrations of cocaine, the formation of cocaethylene, and unique neurochemical effects compared to use of either substance alone or more separated in time.

Distinctions between the neurobiological effects of simultaneous, sequential, and concurrent use of cocaine and alcohol are a grossly understudied area. The few studies that have evaluated this topic, including those by our group (Stennett and Knackstedt, 2020; Stennett et al., 2020), and others (Czoty et al., 2018; Say et al., 2022), find that sequential cocaine-alcohol intake alters glutamate and dopamine homeostasis in the striatum of rats and monkeys, respectively. However, these studies used patterns of polysubstance use (e.g., cocaine before alcohol) that have not yet been confirmed to occur in the real world. To address this gap in knowledge, we developed and tested a new assessment to capture more granular (hourly) temporal patterns of cocaine polysubstance use (Fitzgerald et al., 2022). Here, we use this assessment to identify the most representative temporal patterns of cocaine, alcohol, and cannabis use combinations. To do so, we employed sequence analysis, a method for categorizing patterns in temporally ordered data, and hierarchical clustering.

2. Material and methods

2.1. Sample

Individuals (n=148) recruited from the community were eligible to participate if they 1) were between 18 and 65 years of age, 2) reported any past 30-day cocaine use as well as more than five days of cocaine use within the past 12 months, and 3) reported any past 30-day alcohol and/or cannabis use. Cross-sectional interviews were conducted between December 2019 and March 2021 (the first four months were in-person, then over Zoom due to COVID-19). Recruitment took place via flyers, online advertisements, ResearchMatch, Consent2Share, and through HealthStreet, the University of Florida Clinical and Translational Science Institute's community engagement program. All study protocols were approved by the University of Florida Institutional Review Board.

2.2. Assessments and procedures

Cocaine polysubstance use patterns were assessed with the Polysubstance Use–Temporal Patterns Section (PSU-TPS), developed during the first phase of the study and found to have good-to-excellent test-retest reliability (Fitzgerald et al., 2022). The PSU-TPS interview determined on which of the past 30 days cocaine, alcohol, and cannabis were consumed on the same day or separately. For each pattern of substance use (cocaine+alcohol, cocaine+cannabis, cocaine+alcohol+cannabis), individuals were asked to describe their "typical day" of use by indicating on a 24-hour timeline (from 6 am to 5 am) the hours in which they started and stopped using cocaine, alcohol, and/or cannabis, with up to three episodes (hourly blocks of continuous use) for each substance. For alcohol, quantity of use was reported in standard drinks. For cocaine and cannabis, route of administration and quantity of use were queried, with quantity based on their preferred choice of one of 25 common units (e.g., grams, lines, joints). In addition to the PSU-TPS, the Substance Abuse Module (SAM), a structured and computerized diagnostic interview used internationally (Cottler et al., 1989; Horton et al., 2000), was administered to assess past 12-month DSM-5 cocaine, alcohol, and cannabis use disorders and sociodemographic variables including age, sex, race/ethnicity, and others listed in Table 1. Participants were remunerated $25 USD for time and effort, with a $5 bonus for on-time attendance to their one-time cross-sectional interview.

2.3. Analyses

Among the 148 individuals surveyed, 7 reported no use of alcohol or cannabis on the same day as cocaine in the past 30 days and were excluded. Of the remaining 141 individuals, 104 reported one combination of cocaine polysubstance use (one typical day); 35 reported two combinations (two typical days); and two reported all three combinations (three typical days), resulting in a total of 180 days involving cocaine polysubstance use. Categorical time-series data were converted to state-sequence format, in which each combination of use reported for any hour was represented as a separate categorical state (Figure S1). Based on all possible combinations, eight possible states were defined for each hour: 1) cocaine+alcohol+cannabis co-use (CAM—note that cannabis is referred to as marijuana [M] to differentiate cocaine and cannabis); 2) cocaine+alcohol co-use, no cannabis use (CA); 3) cocaine+cannabis co-use, no alcohol use (CM); 4) alcohol+cannabis co-use, no cocaine use (AM); 5) cocaine-only use; 6) alcohol-only use; 7) cannabis-only use; or 8) no use. To identify the most common patterns of polysubstance use, we used hierarchical clustering, which first required computing a dissimilarity matrix to measure the similarity between pairs of sequences. The optimal matching algorithm determined pairwise similarity, which defines the distance between two sequences as the minimal total "cost," measured by the number of operations (insertions, deletions, and substitutions) required to transform one sequence into another (Studer and Ritschard, 2016). In this context, it accounted for differences in both duration and order of hours of substance use categories observed.

The existence of clusters within the sequence data was confirmed with the Hopkins statistic (H=0.86) (Banerjee and Dave, 2004). Agglomerative hierarchical clustering of the sequences' dissimilarity scores (the similarity between days of polysubstance use) was then conducted using the Ward method (Ward, 1963) to minimize residual variance, generating a dendrogram graph specific to our data (Figure S2). To select an optimal number of polysubstance use patterns, we compared cluster quality indices (Table S1), the size of the estimated clusters, the stability of the estimated clusters (Table S2), and the interpretability of the cluster groups. The average silhouette width (ASW) was chosen as an optimal measure of cluster quality, for its ability to capture both between-cluster separation and within-cluster homogeneity (Studer, 2021). Next, a pseudo-R^2 was computed to quantify the share of sequence discrepancy explained by the clustering solution, and a pseudo-F statistic was calculated that measured clustering significance (Studer et al., 2011). Using the Studer (2021) methodology, sequence analysis clusters were also validated with parametric bootstrapping. To summarize the identified patterns, we also extracted a representative set of sequences for each cluster, defined as a set of non-redundant

Table 1
Characteristics of individuals (n=141) by polysubstance use pattern membership.

	Total (n=141)[a]	Pattern 1: Limited C or CA (n=80)	Pattern 2: Extensive M then CAM (n=34)	Pattern 3: Limited A or M then CA (n=24)	Pattern 4: Extensive CM then CAM (n=7)	Pattern 5: Extensive C then CA (n=13)	p[b]
Age in years, mean (*SD*)	33.9 (14.5)	35.8 (15.6)	34.6 (13.6)	34.3 (14.5)	25.4 (11.8)	27.2 (11.8)	
Female, %	42.6	43.8	47.1	33.3	57.1	38.5	
Race/ethnicity, %							
Hispanic	14.9	15.0	11.8	8.3	14.3	15.4	
Non-Hispanic Black	22.7	28.8	23.5	25.0	14.3	0.0	
Non-Hispanic Other[c]	7.1	7.5	8.8	12.5	14.3	0.0	
Non-Hispanic White	55.3	48.8	55.9	54.2	57.1	84.6	
Months worked in past 12, mean (*SD*)	5.9 (4.5)	6.1 (4.6)	5.7 (4.0)	4.3 (4.4)	7.6 (5.4)	5.8 (4.0)	
Education, %							
≤High school	26.2	26.3	26.5	41.7	28.6	23.1	
>High school	73.8	73.8	73.5	58.3	71.4	76.9	
Marital status, %							
Never married	79.4	82.5	64.7	87.5	85.7	84.6	
Divorced, widowed, or separated	16.3	12.5	32.4	12.5	0.0	15.4	
Married	4.3	5.0	2.9	0.0	14.3	0.0	
Type of cocaine use in lifetime, %							.769
Used both powder and crack cocaine	24.8	25.0	29.4	29.2	14.3	15.4	
Used only powder cocaine	68.8	65.0	67.7	70.8	85.7	76.9	
Used only crack cocaine	6.4	10.0	2.9	0.0	0.0	7.7	
Past 12-month SUD, %							
Cocaine use disorder	50.3	52.5	55.9	62.5	42.9	38.5	.648
Alcohol use disorder	53.2	47.5	50.0	54.2	71.4	84.6	.262
Cannabis use disorder	44.7	38.8	61.8	54.2	71.4	38.5	.138
Number of DSM-5 criteria items endorsed for SUD in past 12 months, mean (*SD*)							
Cocaine use disorder	3.7 (3.3)	3.7 (3.4)	4.9 (3.4)	3.6 (3.5)	2.0 (1.6)	3.5 (2.7)	.208
Alcohol use disorder	4.0 (2.9)	3.7 (2.9)	3.9 (3.1)	4.7 (3.0)	3.7 (3.3)	5.7 (2.3)	.213
Cannabis use disorder	2.9 (2.7)	2.6 (2.6)	3.9 (2.3)	3.1 (3.3)	3.0 (1.7)	2.5 (2.6)	.198
Ever been in drug or alcohol treatment, %	33.3	32.9	39.4	33.3	14.3	30.8	.808
Days used in past 30, mean (*SD*)							
Total days involving any cocaine use	5.7 (6.6)	6.6 (7.5)	6.6 (7.2)	5.9 (6.4)	1.9 (1.5)	5.9 (7.8)	.571
Total days involving any alcohol use	10.9 (9.4)	10.0 (9.4)	9.8 (8.9)	14.2 (9.4)	9.0 (6.2)	15.9 (10.0)	.083
Total days involving any cannabis use	15.3 (12.4)	12.9 (12.2)	24.1 (9.6)	13.4 (11.2)	15.3 (12.6)	9.2 (10.9)	**<.001**
Total days involving any use (cocaine, alcohol, and/or cannabis)	21.2 (9.1)	19.6 (9.4)	26.2 (7.0)	21.1 (8.0)	18.0 (11.2)	19.5 (9.9)	**.006**
Total days involving cocaine-only use	0.8 (2.9)	1.4 (3.8)	0.0	0.2 (0.7)	0.0	0.2 (0.4)	.083
Total days involving alcohol-only use	3.5 (6.5)	3.5 (6.0)	1.7 (5.0)	4.6 (7.7)	2.0 (3.2)	7.9 (8.8)	**.038**
Total days involving cannabis-only use	8.4 (9.9)	6.7 (9.5)	13.5 (10.5)	6.5 (8.1)	8.7 (10.1)	3.2 (5.7)	**.002**
Total days involving CA use	1.6 (2.9)	1.9 (2.8)	0.4 (1.2)	3.0 (4.2)	0.7 (1.5)	2.2 (2.2)	**.007**
Total days involving CM use	1.2 (3.3)	1.6 (4.0)	2.9 (5.8)	0.3 (0.7)	0.3 (0.5)	0.2 (0.6)	.082
Total days involving CAM use	2.1 (4.4)	1.8 (4.2)	3.4 (4.3)	2.4 (4.5)	0.9 (0.7)	3.4 (7.7)	.371

[a] n=17 individuals grouped into more than one cluster; ns for Patterns 1 through 5 add up to 158, but only the 141 individuals are included in the total column. [b] Mixed-effects models (cluster membership as fixed effect and participant ID as random effect). [c] Non-Hispanic Other includes Asian and American Indian or Alaska Native. M: cannabis-only use; A: alcohol-only use; CA: cocaine+alcohol co-use; CM: cocaine+cannabis co-use; CAM: cocaine+alcohol+cannabis co-use; SUD: substance use disorder; SD: standard deviation.

sequences that covered at least 25% of the spectrum of the cluster's sequences (Gabadinho et al., 2011). Representativeness was based on sequences' neighborhood density scores, and redundant sequences were identified as those located within a neighborhood radius of 10% of the maximum theoretical distance of the candidate sequence based on dissimilarity values (Gabadinho et al., 2011).

Individual-level characteristics (sociodemographic variables; past 12-month DSM-5 cocaine, alcohol, and cannabis use disorders; frequency of past 30-day use) and day-level characteristics (routes of administration; quantity of substance use; other substance use reported) were compared between patterns. To account for individuals with more than one category of typical day, mixed-effect models compared clusters across individual- and day-level characteristics. Data were managed using SAS (version 9.4), while sequence and clustering analyses used the 'TraMineR' (Gabadinho et al., 2011), 'cluster' (Maechler et al., 2013), and 'WeightedCluster' packages in RStudio (version 4.1.0). Pattern characterization with mixed-effects models were computed in SAS using the GLIMMIX procedure.

3. Results

3.1. Sample characteristics

Characteristics of the 141 individuals who reported ≥1 day of cocaine co-used with alcohol and/or cannabis in the past 30 days are presented in Table 1. The mean age of individuals was 33.9 years (*SD*=14.5, range 18–65); the majority were male (57.5%), non-Hispanic white (55.3%), completed some college education (73.8%), and never married (79.4%). Over two-thirds reported lifetime use of powder cocaine only (68.8%), with 24.8% reporting both powder and crack cocaine use. Over half met DSM-5 criteria for past 12-month alcohol (53.2%) or cocaine (50.3%) use disorder, while 47.5% met criteria for cannabis use disorder. One-third (33.3%) reported having attended a drug or alcohol treatment program. On average, individuals reported 5.7 cocaine use days (*SD*=6.6), 10.9 alcohol use days (*SD*=9.4), and 15.3 cannabis use days (*SD*=12.4) in the past 30 days.

3.2. Sequence analysis and clustering solution

While the highest ASW value was associated with the three-cluster model (ASW=0.38), the results from our validation analysis (Figure S3) suggested a five-cluster solution (ASW=0.30) was outside the null range. Despite the smaller size of clusters identified, cluster-wise indices confirmed that the five-cluster solution showed the highest between-cluster separation and within-cluster homogeneity (ASW values=0.56, Table S1) as well as high stability (Jaccard index values 0.89, Table S2) and were visibly distinct and interpretable. Moreover, the five-cluster solution was associated with a significant pseudo-*F* (23.39, $p<.001$), with cluster membership explaining 34.8% of sequence discrepancy, and was therefore selected as the final solution. Of the 37 individuals with more than one day represented in the dataset, 20 had all days grouped into the same cluster, while 17 had days grouped into more than one cluster. There were no differences in sociodemographic or substance-related characteristics between those with more than one day belonging to one versus multiple clusters (Table S3).

3.3. Patterns of cocaine polysubstance use

Fig. 1 illustrates the typical days by pattern; the aggregate frequencies of combinations of use at each hour for each pattern (first two columns), and mean duration of drug combinations for each pattern (last column). Pattern 1 was comprised of 96 sequences from 80 individuals and was formed largely by days with no use until the late afternoon or evening, with an average 17.4 hours (*SD*=3.0) of no use, 1.6 hours (*SD*=2.5) of cocaine-only use, and 1.5 hours (*SD*=1.9) of CA co-use

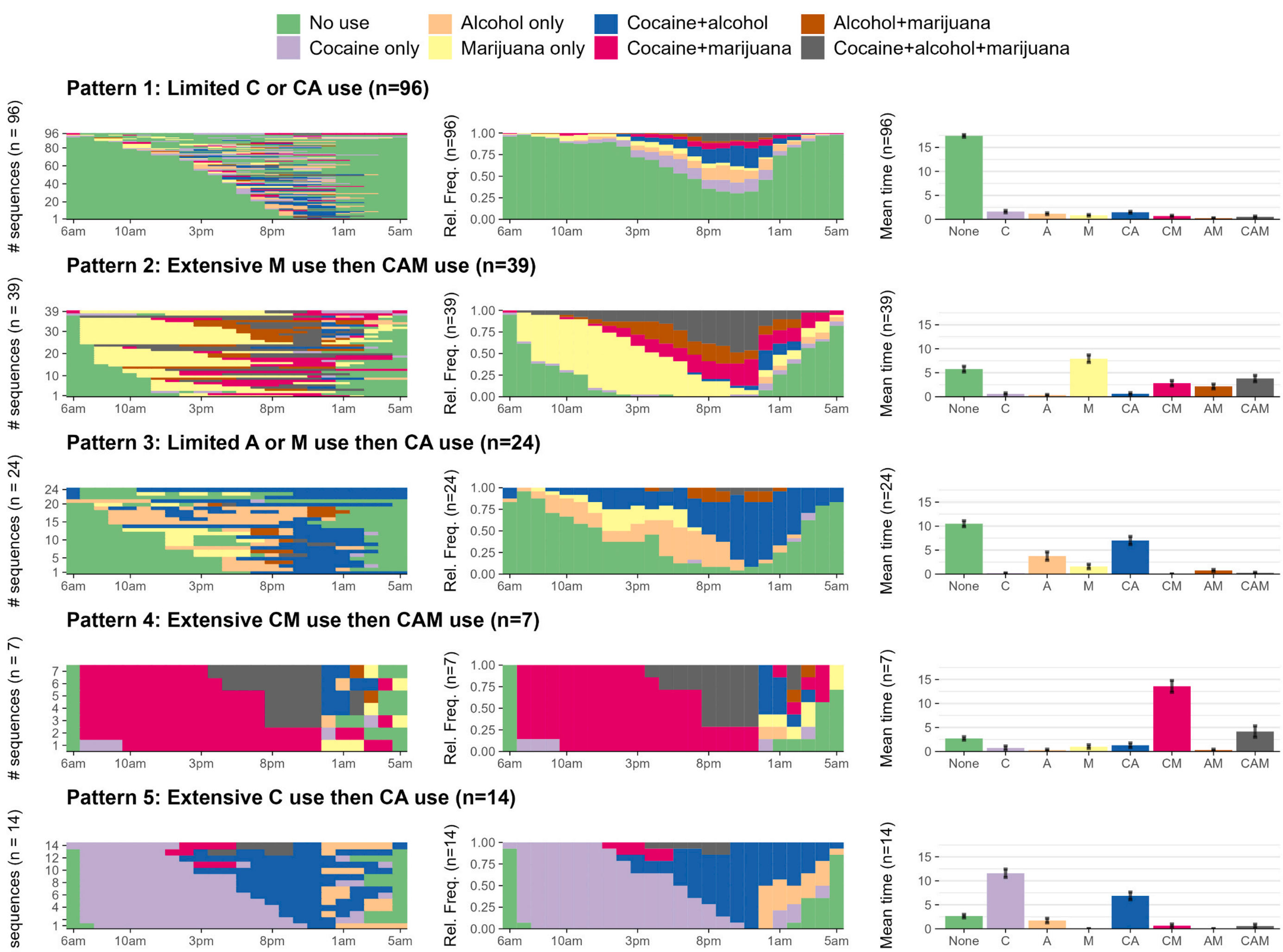

Fig. 1. Plots displaying characteristics of the five patterns of cocaine polysubstance use. Column 1 (from left to right): index plots for typical days of cocaine polysubstance use grouped into each pattern. Column 2: transversal state distribution plots of typical days of polysubstance use in five patterns, with frequencies of the combinations of use at each hour of the typical day plotted over 24 hours. Column 3: mean time (in hours) spent in each state of substance use for the typical days grouped into each pattern.

(limited cocaine/CA, P1). Pattern 2, which included 39 sequences from 34 individuals, involved extensive morning cannabis-only use followed by CAM co-use, with an average 7.9 hours (*SD*=4.9) of cannabis-only use and 3.8 hours (*SD*=4.2) of CAM co-use (extensive cannabis then CAM, P2). Pattern 3, with 24 sequences from 24 individuals, consisted of limited alcohol- or cannabis-only use then CA co-use, with an average 3.8 hours (*SD*=4.3) of alcohol-only use, 1.6 hours (*SD*=2.3) of cannabis-only use, and 7.0 hours (*SD*=4.2) of CA co-use (limited alcohol/cannabis then CA, P3). Finally, though smaller in size, two patterns captured distinct combinations of use. Pattern 4, which included 7 sequences from 7 individuals, involved extensive CM co-use (mean 13.6 hours, *SD*=3.3) followed by CAM co-use (mean 4.1 hours, *SD*=3.1) (extensive CM then CAM, P4). Pattern 5 included 14 sequences from 13 individuals and involved extensive cocaine-only use (mean 11.6 hours, *SD*=3.3) followed by CA co-use (mean 6.9 hours, *SD*=3.0) (extensive cocaine then CA, P5). In summary, both P1 and P3 had the highest frequencies of no use reported in the morning and early afternoon, while P2, P4, and P5 were predominantly characterized by use beginning in the early morning: over half (53.8%) of P2's sequences involved use beginning at 8 am, whereas all sequences in P5 and 94.9% of sequences in P4 involved use beginning at 7 am.

Within each pattern, we extracted the most "typical" days representative of at least 25% of the days in the cluster (Fig. 2). From the representative sets of sequences, measures of order and duration are summarized for each cluster. P1 was best represented by 2–5 hours of evening use of cocaine and alcohol, where cocaine use began either at the same time as alcohol use or was preceded and followed by 1–2 hours of alcohol-only use. For P2, which showed the greatest discrepancy within representatives' assigned sequences, all representative days began with morning or early afternoon cannabis use; the majority began with cannabis-only use, though all involved cocaine polysubstance use (either CM or CAM) at night. Representative sequences for P3 also showed a degree of heterogeneity: while all involved multiple hours of CA co-use, which mostly began at night, three representative sequences involved short (2–4 hour) episodes of cannabis-only use in the morning, while one began with alcohol-only use. Only one sequence was selected to represent P4, which consisted of 13 hours of CM co-use from early morning to evening, followed by four hours of CAM co-use and two hours of CA co-use, ending with cannabis-only use. For P5, both of two representative sequences began with cocaine-only use at 7 am, lasting until transitioning into CA co-use at night.

3.4. Additional characterization of patterns

Table 1 displays sociodemographic and substance-related differences among individuals by pattern membership. Due to cell sparseness, sociodemographic comparisons were not included. Patterns varied according to frequency of past 30-day substance use, with individuals in

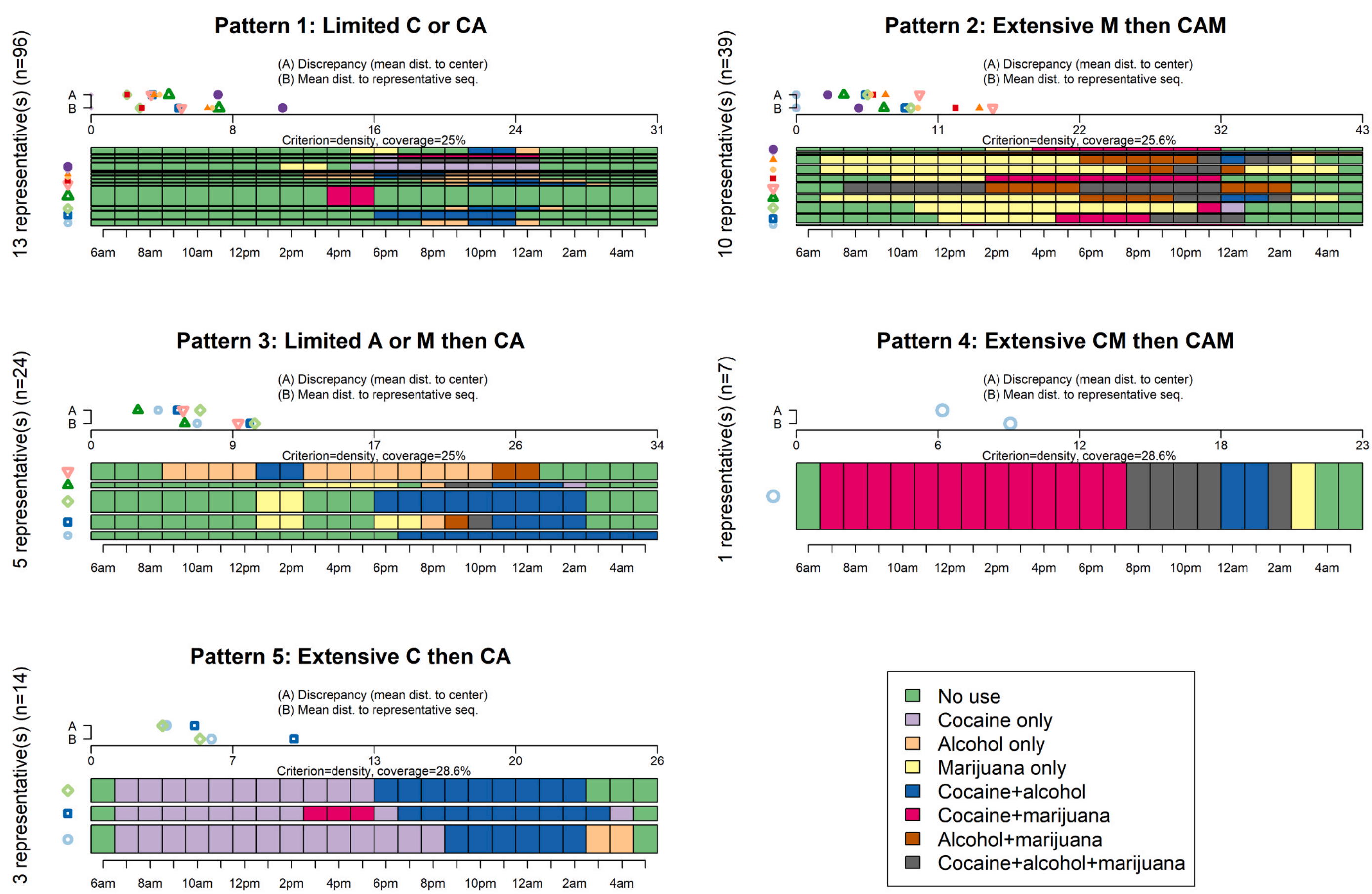

Fig. 2. Representative sets of typical days of polysubstance use in five patterns. Sequences (days) in each cluster were first sorted in descending order according to their neighborhood density scores (the number of sequences in the neighborhood of candidate each sequence), and representative sets were selected by successively removing "redundant" sequences from this list (those located within a neighborhood radius of 10% of the maximum theoretical distance of the candidate sequence based on dissimilarity values). Each redundant sequence was then assigned to its closest representative. In this figure, sequences are plotted from bottom-up based on representativeness, with the vertical width of each sequence corresponding to its number of assigned redundant sequences. The threshold used for each representative set was 25% coverage (the set of representatives which covered at least 25% of the sequences within the given cluster). The quality of the representative set of sequences for each pattern is indicated by (A) the discrepancy within the representative set's assigned sequences and (B) the mean distance between the representative set and its assigned sequences.

the extensive cannabis then CAM (P2) pattern reporting significantly greater mean days of cannabis (24.1, SD=9.6) and any substance use (26.2, SD=7.0) compared to those grouped into other patterns (p<.001 and p=.006, respectively). Those in the extensive cannabis then CAM (P2) pattern also reported significantly fewer mean days of CA use in the past 30 days (0.4, SD=1.2) compared to those in P1 (p=.010), P3 (p=.001), and P5 (p=.040), and significantly greater mean days of cannabis-only use in the past 30 days (13.5, SD=10.5) compared to those in P1 (p=.001), P3 (p=.005), and P5 (p=.001). Individuals in the extensive cocaine then CA (P5) pattern reported significantly greater mean days of alcohol-only use in the past 30 days (7.9, SD=8.8) compared to those in P1 (p=.020) and P3 (p=.003). Considering characteristics of the sequence-level data (Table 2), days in P5 involved a significantly greater quantity of crack/cocaine use (34.0 rocks) than other patterns (p=.015), without a difference in mean grams of cocaine consumed. Days in P1 and P2 involved significantly fewer drinks on average versus P3, P4, and P5, while days in P2 involved a significantly greater mean quantity of grams of cannabis used (6.0, SD=4.5) compared to those in P1 and P3. Notably, a proportion of days (<15.5%) grouped in every pattern but the extensive CM then CAM (P4) pattern also involved the use of stimulants, opioids, and/or sedatives.

4. Discussion

The present analysis applied sequence analysis with hierarchical clustering to identify distinct patterns of hour-level polysubstance use among adults who used cocaine with alcohol and/or cannabis. Overall, individuals' typical days of cocaine polysubstance use fit one of five distinct patterns, characterized by combinations of substance use and the timing, order, duration, and quantity of episode-level substance use. These findings are significant because we have identified the preferred timing and ordering of these drugs, which could have implications for the pharmacokinetics and pharmacodynamics of these substances.

These findings provide the most detailed, quantitative examination to date of the temporality of cocaine polysubstance use. Previous studies assessed whether cocaine preceded or followed alcohol use, finding both that alcohol preceded cocaine use on days in which both were used (Barrett et al., 2006, 2005; Macdonald et al., 2014; Staines et al., 2001) and vice versa (Gossop et al., 2006). Our more granular interview identified simultaneous CA co-use as the most common pattern (P1), with the same start times reported for both cocaine and alcohol use. Although it is possible that alcohol was consumed before cocaine within the same hour (or vice versa), the hour-level detail provides greater resolution than studies that only enumerated the order in which they were used. The less-prevalent pattern of extensive cocaine then CA involved sequential polysubstance use, with cocaine-only use preceding simultaneous CA co-use over the course of almost 24 hours. The prevalence of simultaneous cocaine and alcohol use reported in the present study is in agreement with the prevalence of cocaethylene detection in individuals seeking treatment for acute cocaine intoxication (Zucoloto et al., 2021).

To date, only one study has examined temporal patterns of cocaine and cannabis use, and none has evaluated neurobiological changes produced by co-use. Barrett and colleagues (2006) found that 42.9% of a sample of university students used cannabis before cocaine during their most recently recalled use episode, whereas 37.5% used cannabis over multiple sessions with cocaine use. Similar to those findings, the most representative sequence for the extensive cannabis then CAM (P2) pattern in the current study began with cannabis-only use before cocaine use, while the most representative sequence for the extensive CM then CAM (P4) pattern began with simultaneous CM co-use. Some evidence indicates that delta-9-tetrahydrocannabinol (THC, the primary psychoactive component of cannabis) enhances brain levels of subsequently administered cocaine, suggesting that cannabis use could potentiate the psychoactive effects of cocaine (Lukas et al., 1994; Reid and Bornheim, 2001). Of note, almost all representative days involving sequential polysubstance use began with a single drug (e.g., cocaine-, alcohol-, or cannabis-only) before transitioning to polysubstance use (e.g., CA, CM, or CAM). Transitions from a first to a second single drug were rare.

Beyond order of use, timing was also a significant factor that differentiated the five patterns. When cocaine and alcohol use were reported, the CA patterns predominantly involved late afternoon or

Table 2
Characteristics of day-level data (n=180) by polysubstance use pattern membership.

	Total (n=180)	Pattern 1: Limited C or CA (n=96)	Pattern 2: Extensive M then CAM (n=39)	Pattern 3: Limited A or M then CA (n=24)	Pattern 4: Extensive CM then CAM (n=7)	Pattern 5: Extensive C then CA (n=14)	p[a]
Routes of cocaine administration on typical day, %							
Snorted or sniffed	76.1	72.9	76.9	79.2	100.0	78.6	–[b]
Smoked	30.0	30.2	38.5	25.0	0.0	28.6	–[b]
Oral	8.9	7.3	5.1	16.7	14.3	14.3	.673
Routes of cannabis administration on typical day, %							
Smoked	82.2	76.0	100.0	83.3	100.0	64.3	–[b]
Oral	7.2	7.3	10.3	4.2	0.0	7.1	–[b]
Vaped	5.6	6.3	2.6	12.5	0.0	0.0	–[b]
Quantity of cocaine use on typical day, mean (*SD*)							
Number of grams (n=87)	2.7 (6.9)	3.6 (9.4)	1.9 (2.3)	1.3 (1.0)	1.2 (0.0)	1.6 (0.8)	.652
Number of lines (n=61)	4.8 (3.6)	4.9 (3.6)	3.2 (2.3)	6.1 (4.1)	4.4 (4.9)	5.1 (3.9)	.669
Number of rocks (n=21)	12.1 (12.6)	6.7 (5.2)	16.4 (17.1)	7.0 (4.2)	0.0	34.0 (0.0)	**.015**
Quantity of alcohol use on typical day, mean (*SD*)							
Number of drinks (n=165)	12.1 (10.4)	9.6 (8.1)	9.7 (8.1)	18.3 (11.8)	18.5 (19.1)	20.3 (13.0)	**<.001**
Quantity of cannabis use on typical day, mean (*SD*)							
Number of grams (n=58)	3.3 (3.6)	2.0 (2.4)	6.0 (4.5)	2.2 (1.7)	0.0	1.0 (1.4)	**<.001**
Number of joints (n=53)	2.6 (2.0)	2.3 (1.5)	3.7 (2.8)	3.3 (2.7)	2.0 (1.0)	1.7 (1.0)	.113
Number of blunts (n=34)	5.9 (5.3)	5.6 (4.1)	9.2 (8.9)	3.0 (3.6)	1.0 (0.0)	0.0	.255
Number of hits (n=26)	8.5 (6.7)	7.7 (3.4)	11.0 (10.6)	7.7 (4.9)	10.3 (12.1)	5.7 (4.2)	.769
Other substances used on typical day, %							
Stimulants	7.2	5.2	15.4	4.2	0.0	7.1	–[b]
Opioids	6.1	6.3	10.3	0.0	0.0	7.1	–[b]
Sedatives, tranquilizers, or benzodiazepines	5.6	2.1	7.7	12.5	0.0	14.3	–[b]

[a] Mixed-effects models (cluster membership as fixed effect and participant ID as random effect). [b] Model did not converge. M: cannabis-only use; A: alcohol-only use; CA: cocaine+alcohol co-use; CM: cocaine+cannabis co-use; CAM: cocaine+alcohol+cannabis co-use; SD: standard deviation.

evening use, whereas the CM patterns involved more early morning use and use throughout the day. Both CM patterns (P2 and P4) began with cannabis use in the morning (either cannabis-only or CM), consistent with prevalence of morning use among persons who use cannabis frequently (Earleywine et al., 2016; Gunn et al., 2021), although this has not been described previously in relation to timing of cocaine use. However, our interview did not ask what time participants woke up on their "typical day," so limited conclusions can be drawn at this time regarding circadian patterns of drug use.

Despite considerable differences in hour-by-hour patterns of polysubstance use, the five clusters identified were similar regarding number of days in a month with cocaine use (mean of 5.7). This intermittent pattern of cocaine use has been reported in epidemiological studies and observed in the demographic tables of most clinical trials for cocaine use disorder treatments (Simon et al., 2002), with ranges of 2–16 days/month of cocaine use reported, depending on the study (Mariani et al., 2021; Mongeau-Pérusse et al., 2022; Rizkallah et al., 2022; Schmitz et al., 2021). Not all participants met criteria for cocaine use disorder, and the somewhat lower average days of use per month may be related to this feature. Total days of alcohol use in the past 30 days did not differ between clusters (mean of 10.9), whereas days with cannabis use did (mean of 15.2, range of 9.2–24.1). Although alcohol and cannabis were used on a greater number of days per month than cocaine, neither were used daily. In contrast, most preclinical research employs daily drug administration sessions, and thus the neurobiological changes revealed may not reflect those found in most humans, as the duration of withdrawal can have as large an impact on brain neurochemistry as drug intake itself (Miguéns et al., 2008; Smaga et al., 2020). Thus, the pattern of polysubstance use over day-to-day timescales could have a significant impact on neurobiological mechanisms supporting substance use.

A key finding of the present work is that, in addition to a lack of effect of polysubstance use pattern on the number of days of cocaine use per month, there were also no differences in the amount of cocaine used per day, nor in the prevalence of cocaine use disorder across patterns. There was an effect of pattern on the number of rocks of cocaine used during episodes of cocaine use, indicating that some clusters may uniquely use more of this form of cocaine via the inhaled route. The route of administration impacts cocaine pharmacokinetics, with the intranasal route exhibiting reduced plasma cocaine concentrations relative to inhalation (Cone, 1995). However, the prevalence of each route of cocaine administration did not differ between the five clusters in the present study, indicating that members of each cluster employed various routes of administration. Overall, while our findings suggest that hourly sequences of cocaine polysubstance use are not strongly associated with substance use disorders (SUDs), it should also be noted that measurements of polysubstance use patterns were assessed over the past 30-day period, while the measurement of SUDs considered the past 12 months. Future studies should consider exploring whether other aspects of polysubstance use patterns, such as frequency of polysubstance use, are more strongly associated with risk of SUDs.

4.1. Limitations

Data on detailed drug use patterns were obtained retrospectively during cross-sectional interviews. While the SAM and PSU-TPS interviews have demonstrated good test-retest reliability (Cottler et al., 1989; Fitzgerald et al., 2022; Horton et al., 2000), there is a possibility for measurement error from self-reported recall. Future research should compare these findings to studies using prospective methods, such as ecological momentary assessment. Individuals were also given the option of reporting quantity of cocaine and cannabis use in up to 25 different units. While this flexibility allows participants to report quantity of use based on their preferred unit, variations in reported units affect standardization and comparison of quantities of cocaine and cannabis, which limited analyses to stratification of quantities of both substances by the most commonly reported units. Additionally, since the PSU-TPS assessment collected start and end times for substance use episodes at the hour-level, we were unable to consider within-hour orders of substance use administration. As the data collection period included the COVID-19 pandemic and associated restrictions, the interview modality also changed to remote interviews. The pandemic may have impacted substance use patterns, including cocaine polysubstance use, which is beyond the scope of this analysis; however, we found no differences in test-retest reliability in the PSU-TPS by interview modality (Fitzgerald et al., 2022). Finally, the sample was relatively small and primarily recruited from the North Central Florida area and may not be generalizable to other populations.

4.2. Conclusion

In a sample of community-recruited adults, we identified five distinct patterns of cocaine polysubstance use, characterized by timing, order, and duration of use. While these patterns were not accompanied by different amounts of cocaine used or prevalence of cocaine use disorder, the patterns have the potential to produce distinct pharmacokinetic and pharmacodynamic differences, including the production of cocaethylene in the patterns endorsing simultaneous CA use. Future research should validate these findings using other assessment methods, including those which capture other substances co-administered with cocaine beyond alcohol and cannabis, and examine other dimensions of polysubstance use related to typical days of use, such as how patterns of use vary over time by context and setting. To develop effective treatments for cocaine use disorder, clinical studies must consider the effects of polysubstance use, including potential effects of cocaethylene. Our characterization of human cocaine-alcohol-cannabis polysubstance use represents an important step toward building preclinical models that capture real-world neurobiological consequences of polysubstance use.

Preclinical models of SUDs have recently undergone revisions to become more translational, as seen in the adoption of "long-access" paradigms that use 6–8 hours per day of drug self-administration and "intermittent" access paradigms that provide limited access to drug self-administration, interspersed with periods when the drug is not available within a several-hour-long self-administration session. Both models produce alterations in neurobiology (particularly in dopamine signaling) relative to "standard" models involving 2 hours of continuous drug access (Carr et al., 2020). Interestingly, while these models have urged the field to consider the importance of temporal aspects of drug availability, they are not based on real-world data. The current data indicate most persons who use cocaine report use for periods shorter than 6 hours, and only 1–2 days/week on average. We posit that even more detailed epidemiological assessments of substance use are necessary for the preclinical field, of which some of us are members, to build animal models of substance use that reflect real-world patterns of human behavior. The present manuscript describes such patterns that can be used to inform such models, including those of polysubstance use, in a data-driven manner.

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CRediT authorship contribution statement

Linda B. Cottler: Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization. **Anna Wang:** Writing – review & editing, Project administration. **Catherine W. Striley:** Writing – review & editing, Investigation, Funding acquisition, Conceptualization. **Barry Setlow:** Writing – review

& editing, Funding acquisition, Conceptualization. **Lori Knackstedt:** Writing – review & editing, Funding acquisition, Conceptualization. **Nicole D. Fitzgerald:** Writing – review & editing, Writing – original draft, Visualization, Project administration, Investigation, Formal analysis. **Yiyang Liu:** Writing – review & editing, Project administration, Investigation.

Declaration of Competing Interest

The authors have no potential conflicts to declare.

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Contributors

All authors are responsible for this reported research. L.B. Cottler, L. Knackstedt, B. Setlow, and C.W. Striley conceptualized and designed the study; N.D. Fitzgerald, Y. Liu, A. Wang, C.W. Striley, and L.B. Cottler performed the research; N.D. Fitzgerald conducted the statistical analyses and drafted the manuscript, and all authors interpreted results and critically reviewed and revised the manuscript. All authors approved the final manuscript as submitted.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.drugalcdep.2024.111272.

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